Implicit differentiation

$$x^3+y^3=6xy \qquad \frac{dy}{dx}$$

$$g_1(x) = x^3+y^3(x) \qquad g_1'(x) = 3x^2+3y^2y'$$
$$g_2(x) = 6xy(x) \qquad g_2'(x) = 6xy'+6y$$

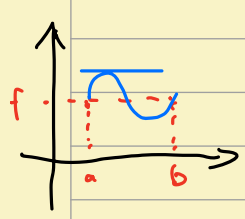
$$\Rightarrow \quad 3x^2+3y^2y' = 6xy'+6y \quad \Rightarrow \quad y' = \frac{2y-x^2}{y^2-2x}$$

logarithmic differentiation

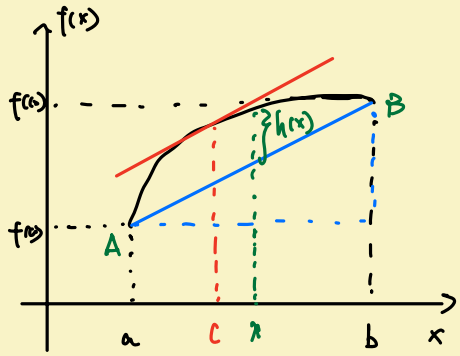
$$y = x^{\sqrt{x}} \qquad (fg)' = f'\cdot g + f\cdot g'$$
$$\Rightarrow \ln y = \ln x^{\sqrt{x}} = \sqrt{x}\cdot \ln x$$
$$y' = x^{\sqrt{x}}\left(\sqrt{x}\cdot\frac{1}{x} + (\ln x)\cdot\frac{1}{2\sqrt{x}}\right)$$

Rolle's theorem $\Rightarrow$ (Lagrangian) mean value theorem
$\Rightarrow$ (Cauchy) mean value theorem
$\Rightarrow$ L'Hospital rule

Rolle's thm : A function f ① continuous on $[a,b]$
② differentiable on (a,b)
③ $f(a) = f(b)$



$\Rightarrow \exists\, c \in (a,b)$ such that $f'(c) = 0$



$$m = \frac{f(b) - f(a)}{b-a}$$

Thm: (The mean value theorem) Suppose f is a function satisfies

① f is continuous on the closed interval $[a, b]$.

② f is differentiable on the open interval (a, b).

Then, there is a number $c \in (a, b)$, such that

$$f'(c) = \frac{f(b) - f(a)}{b-a}$$

Proof: The equation of the secant line AB can be written as

$A(a, f(a))$

$B(b, f(b))$

$$y = f(a) + \frac{f(b) - f(a)}{b-a}(x-a)$$

Define $\quad h(x) = f(x) - f(a) - \frac{f(b) - f(a)}{b-a}(x-a)$

Then we have $h(x)$ is ① continuous on $[a,b]$, and

② differentiable on (a,b).

$$\left.\begin{aligned} ③\ h(a) &= f(a) - f(a) - \frac{f(b) - f(a)}{b-a}(a-a) = 0 \\ h(b) &= \cancel{f(b) - f(a)} - \cancel{\frac{f(b) - f(a)}{b-a}(b-a)} = 0 \end{aligned}\right\} \Rightarrow h(a) = h(b) = 0$$

By Rolle's theorem, there is a point $c \in (a,b)$ such that $h'(c) = 0$

$$f'(c) = \frac{f(b) - f(a)}{b-a}$$

Corollary: Suppose that f is continuous on $[a,b]$ and differentiable on (a,b)

① If $f'(x) \equiv 0$ for all $x \in (a,b)$, then f is constant on $[a,b]$.

② If $f'(x) > 0$ for all $x \in (a,b)$, then f is strictly increasing on $[a,b]$

③ If $f'(x) < 0$ for all $x \in (a,b)$, then f is strictly decreasing on $[a,b]$

Proof

$$\frac{f(b) - f(c)}{b-a} = f'(c) \qquad c \in (a,b)$$

Ex. Show that if $-\frac{\pi}{4} < a < b < \frac{\pi}{4}$, then

$$|\tan b - \tan a| < 2|b-a|$$

$f(x) = \tan x \qquad f'(x) = \sec^2 x \qquad \sec x = \frac{1}{\cos} x$

By the mean value theorem: $\dfrac{\tan b - \tan a}{b-a} = \sec^2 c$

$\sec^2 c = \dfrac{1}{\cos^2 x} \qquad \cos^2 x > \frac{1}{2} \ \forall\ x \in (-\frac{\pi}{4}, \frac{\pi}{4})$

$$\Rightarrow |\tan b - \tan a| < 2|b-a|$$

Corollary. Suppose f is continuous on $(x_0-\varepsilon, x_0+\varepsilon)$, and is differentiable on $(x_0-\varepsilon, x_0) \cup (x_0, x_0+\varepsilon)$, and $\lim\limits_{x\to x_0} f'(x)$ exists

then f is differentiable on x_0, and

$$f'(x_0) = \lim_{x\to x_0} f'(x)$$

Proof: Take $x \in (x_0, x_0+\varepsilon)$, then the mean value theorem holds on $[x_0, x]$, there exists $\xi \in (x_0, x)$ $\Rightarrow$ $\dfrac{f(x)-f(x_0)}{x-x_0} = f'(\xi)$

When $x \to x_0^+$, $\xi \to x_0^+$, then $\lim\limits_{x\to x_0^+} \dfrac{f(x)-f(x_0)}{x-x_0} = \lim\limits_{\xi\to x_0^+} f'(\xi) = f'(x_0+0)$

Similarly, we have $\lim\limits_{\zeta\to x_0^-} f(\zeta) = f'(x_0-0)$

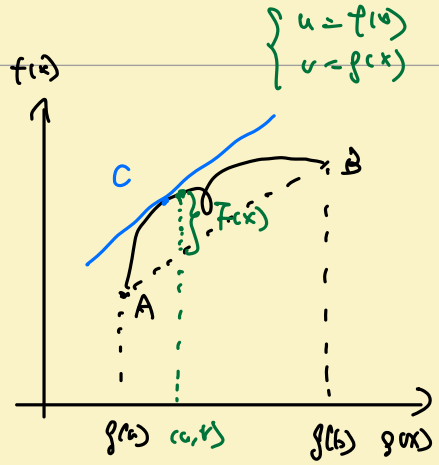
Since $\lim\limits_{x\to x_0} f'(x) = k$ exists, $f'(x_0-0) = f'(x_0+0) = k$.

E.x. $f(x) = \begin{cases} x + \sin x^2 & x \le 0 \\ \ln(1+x) & x > 0 \end{cases} \qquad f'(x) \begin{cases} 1 + 2x\cdot\cos x^2 & x \le 0 \\ \frac{1}{1+x} & x > 0 \end{cases}$

$f'(0-0) = \lim\limits_{x\to 0^-} (1 + 2x\cos x^2) = 1$

$f'(0+0) = \lim\limits_{x\to 0^+} \dfrac{1}{1+x} = 1$

$\Rightarrow f'(0) = 1$



Thm (Cauchy's mean value theorem) Suppose f and g.

① f and g are continuous on $[a, b]$

② f and g are differentiable on (a, b), $g'(x) \neq 0$ $\forall x \in (a, b)$

Then, there exists $c \in (a, b)$, such that

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(c)}{g'(c)}$$

Proof: Define $F(x) = f(x) - f(a) - \frac{f(b) - f(a)}{g(b) - g(a)} (g(x) - g(a))$

By Rolle's theorem $F'(\xi) = f'(\xi) - \frac{f(b) - f(a)}{g(b) - g(a)} g'(\xi) = 0$

$$\Rightarrow \frac{f'(\xi)}{g'(\xi)} = \frac{f(b) - f(a)}{g(b) - g(a)}$$

E.x. Suppose f is differentiable on $(0, 1]$, $\lim_{x \to 0^+} \sqrt{x} f'(x) = A$. Prove that f is uniformly continuous on $(0, 1]$.

Proof: Let $M = |A| + 1$. Since $\lim_{x \to 0^+} \sqrt{x} f'(x) = A$, then $\exists\, 0 < \delta < 1$. when $0 < x < \delta$, $\Rightarrow |\sqrt{x} f'(x)| < M$, for any $x, y \in (0, \delta]$, $x < y$, by Cauchy's mean value theorem, there exists $c \in (x, y)$, such that

$$\left| \frac{f(x) - f(y)}{\sqrt{x} - \sqrt{y}} \right| = 2 |\sqrt{\xi} f'(\xi)| < 2M$$

$$\Rightarrow |f(x) - f(y)| \leq 2M |\sqrt{x} - \sqrt{y}|$$

Since $y = \sqrt{x}$ is uniformly continuous on $(0, \delta]$, then $f(x)$ is uniformly continuous on $(0, \delta]$. Since f is differentiable on $[\delta, 1]$, then f is continuous on $[\delta, 1]$, thereby is uniformly continuous on $[\delta, 1]$.

L'Hospital's rule $\quad \frac{0}{0}, \frac{\infty}{\infty}$

Corollary (L'Hospital's rule). Suppose f and g are differentiable and $g'(x) \neq 0$ on an open interval that contains x_0. Suppose that

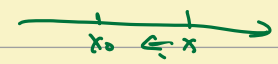
① $\lim_{x \to x_0} f(x) = 0$ and $\lim_{x \to x_0} g(x) = 0$, or

② $\lim_{x \to x_0} f(x) = \infty$ and $\lim_{x \to x_0} g(x) = \infty$.

③ $\lim_{x \to x_0} \frac{f'(x)}{g'(x)} = A \quad (A \in \mathbb{R},\ A = \pm\infty)$

Then $\lim_{x \to x_0} \frac{f(x)}{g(x)} = \lim_{x \to x_0} \frac{f'(x)}{g'(x)} = A$

Proof: $f(x_0) = g(x_0) = 0$



$$\lim_{x \to x_0^+} \frac{f(x) - f(x_0)}{g(x) - g(x_0)} = \lim_{c \to x_0^+} \frac{f'(c)}{g'(c)} \qquad \begin{array}{l} x \to x_0 \\ c \to x_0 \end{array}$$

$$\Rightarrow \lim_{x \to x_0^+} \frac{f(x)}{g(x)} = \lim_{c \to x_0^+} \frac{f'(c)}{g'(c)}$$

E.x. $\lim_{x \to \pi} \frac{1 + \cos x}{\tan^2 x} = \lim_{x \to \pi} \frac{-\sin x}{2 \tan x \sec^2 x} = -\lim_{x \to \pi} \frac{\sin x}{2 \frac{\sin x}{\cos x} \frac{1}{\cos^2 x}}$

$= -\lim_{x \to \pi} \frac{\cos^3 x}{2} = \frac{1}{2}$

$\ln(1+x) \sim x$

$\lim_{x \to 0} \frac{e^x - (1+2x)^{\frac{1}{2}}}{\ln(1+x^2)} = \lim_{x \to 0} \frac{e^x - (1+2x)^{\frac{1}{2}}}{x^2} = \lim_{x \to 0} \frac{e^x - (1+2x)^{-\frac{1}{2}}}{2x}$

$= \lim_{x \to 0} \frac{e^x - (1+2x)^{-\frac{3}{2}}}{2} = 1$

$\lim_{x \to 0^+} \frac{\sqrt{x}}{1 - e^{\sqrt{x}}}$ let $t = \sqrt{x}$ $\quad x \to 0^+ \quad t \to 0^+$

$= \lim_{t \to 0^+} \frac{t}{1 - e^t} = \lim_{t \to 0^+} \frac{1}{-e^t} = -1$

$$\lim_{x\to+\infty} \frac{x+\sin x}{x} = \lim_{x\to\infty}\left(\frac{x}{x}+\frac{\sin x}{x}\right) = 1$$

$$= \lim_{x\to\infty} \frac{1+\cos x}{1} \quad \times$$

E.x. $\lim_{x\to 0^+} x\ln x$ $\qquad 0\cdot\infty \Rightarrow \frac{\infty}{\infty}$

$$= \lim_{x\to 0^+} \frac{\ln x}{\frac{1}{x}} = \lim_{x\to 0^+} \frac{\frac{1}{x}}{-\frac{1}{x^2}} = \lim_{x\to 0^+}(-x) = 0$$

$$\lim_{x\to 0} (\cos x)^{\frac{1}{x^2}} \qquad 1^\infty \Rightarrow e^{\frac{0}{0}} \qquad e^{\ln x} = x$$

$$= \lim_{x\to 0} \exp\left(\frac{1}{x^2}\ln\cos x\right) = \exp\left(\lim_{x\to 0}\frac{\ln\cos x}{x^2}\right) = \exp\left(\lim_{x\to 0}\frac{-\tan x}{2x}\right)$$

$$= e^{-\frac{1}{2}}$$

$$\lim_{x\to 0^+} (\sin x)^{\frac{a}{1+\ln x}} \quad (a\in\mathbb{R}) \qquad 0^0 \Rightarrow e^{\frac{\infty}{\infty}}$$

$$\lim_{x\to 0^+} \frac{a}{1+\ln x}\cdot\ln\sin x = \lim_{x\to 0^+}\frac{k\frac{\cos x}{\sin x}}{\frac{1}{x}} = \lim_{x\to 0^+} k\cos x\cdot\frac{x}{\sin x} = k$$

$$\Rightarrow e^k$$

$$\lim_{x\to+\infty}\left(x+\sqrt{1+x^2}\right)^{\frac{1}{\ln x}} = \exp\left(\lim_{x\to+\infty}\frac{\ln(x+\sqrt{1+x^2})}{\ln x}\right) \qquad \infty^0 \Rightarrow e^{\frac{\infty}{\infty}}$$

$$\lim_{x\to+\infty}\frac{\ln(x+\sqrt{1+x^2})}{\ln x} = \lim_{x\to+\infty}\frac{\frac{1}{x+\sqrt{1+x^2}}\cdot\left(1+x\frac{1}{\sqrt{1+x^2}}\right)}{\frac{1}{x}} = \lim_{x\to+\infty}\frac{\frac{1}{x+\sqrt{1+x^2}}\,\frac{\sqrt{1+x^2}+x}{\sqrt{1+x^2}}}{\frac{1}{x}}$$

$$= \lim_{x\to+\infty}\frac{\frac{1}{\sqrt{1+x^2}}}{\frac{1}{x}} = \lim_{x\to\infty}\frac{x}{\sqrt{1+x^2}} = 1$$

$\infty - \infty \Rightarrow \frac{0}{0}$

$$\lim_{x\to 1}\left(\frac{1}{x-1}-\frac{1}{\ln x}\right)=\lim_{x\to 1}\frac{\ln x - x+1}{(x-1)\ln x}=\lim_{x\to 1}\frac{\frac{1}{x}-1}{\frac{x-1}{x}+\ln x}$$

$$=\lim_{x\to 1}\frac{1-x}{x-1+x\ln x}=\lim_{x\to 1}\frac{-1}{2+\ln x}=-\frac{1}{2}$$

$$\lim_{x\to\frac{\pi}{2}^-}(\sec x-\tan x)=\lim_{x\to\frac{\pi}{2}^-}\left(\frac{1}{\cos x}-\frac{\sin x}{\cos x}\right)=\lim_{x\to\frac{\pi}{2}^-}\frac{1-\sin x}{\cos x}$$

$$=\lim_{x\to\frac{\pi}{2}^-}\frac{-\cos x}{-\sin x}=0$$

E.g. Suppose that f is differentiable $[0,+\infty)$, $\lim_{x\to+\infty}[f(x)+f'(x)]=A$.
Prove $\lim_{x\to+\infty}f(x)=A$.

Proof: $$\lim_{x\to+\infty}f(x)=\lim_{x\to+\infty}\frac{e^x\cdot f(x)}{e^x}=\lim_{x\to+\infty}\frac{e^x f(x)+e^x\cdot f'(x)}{e^x}$$

$$=\lim_{x\to+\infty}[f(x)+f'(x)]=A$$

E.g. Suppose that $f(x)=\begin{cases}\frac{g(x)}{x} & x\neq 0\\ 0 & x=0.\end{cases}$

If $g(0)=g'(0)=0$, $g''(0)=3$. Find $f'(0)$.

Solve: $$\frac{f(x)-f(0)}{x-0}=\frac{f(x)}{x}=\frac{g(x)}{x^2}$$

$$f'(0)=\lim_{x\to 0}\frac{f(x)-f(0)}{x-0}=\lim_{x\to 0}\frac{g(x)}{x^2}=\lim_{x\to 0}\frac{g'(x)}{2x}=\lim_{x\to 0}\frac{g''(x)}{2}=\frac{3}{2}$$

$$\Rightarrow f'(0)=\frac{3}{2}$$